

Presentation on The process of EV Charging

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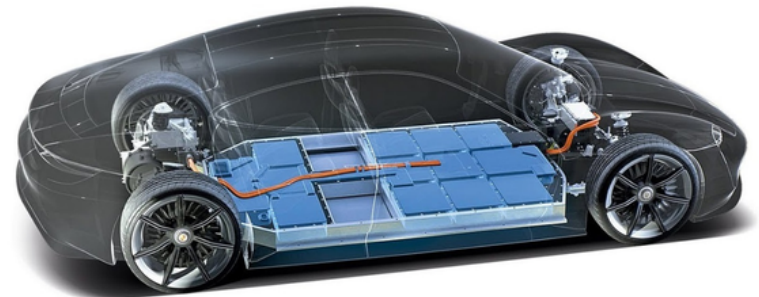
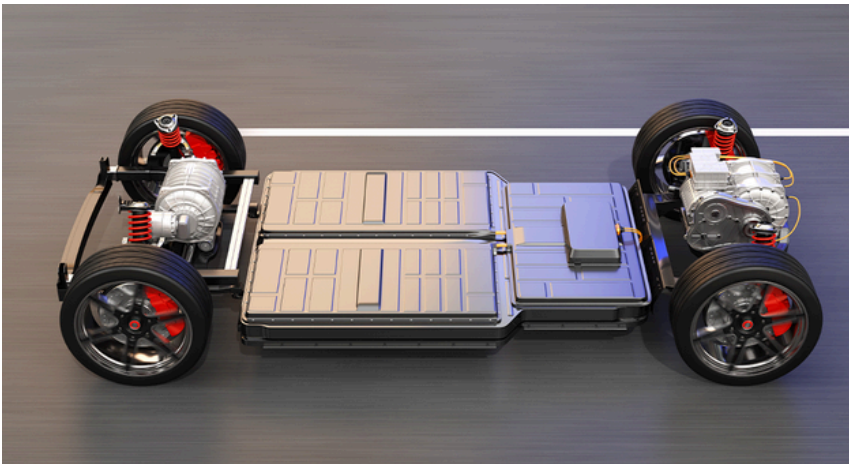
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- What people thinks about the ev chaging
large batteries, single chargers etc
- what ev charging actually is
set of small battery connected in series, lithium can not go above 4.2v
- ev charging station
converting the low voltage to high voltage dc and provide balanced supply buy
comuniaciton to the bms
- ev charging onboard system
how the bms monitor the voltage of each cell and and bypass the extra current ,
why we can not seperatly charge each battery

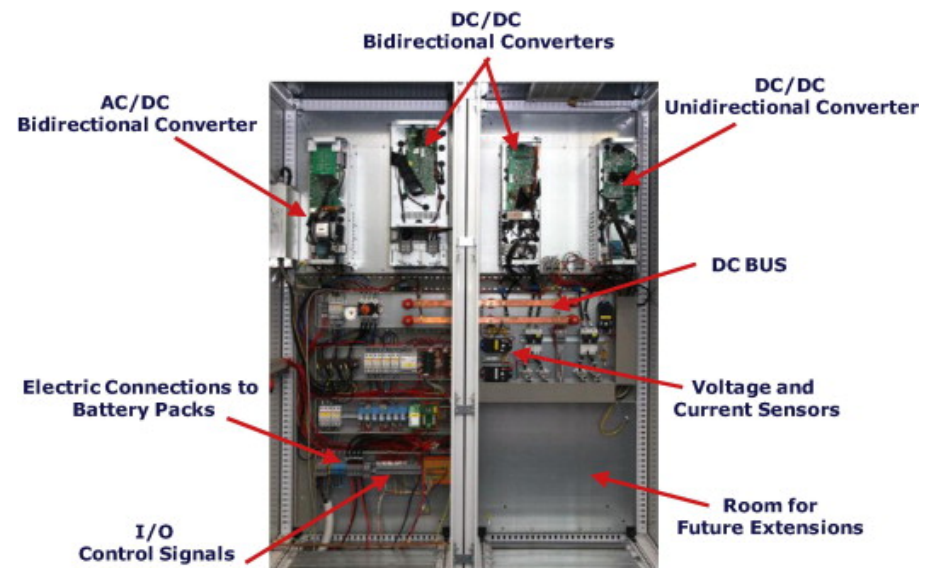
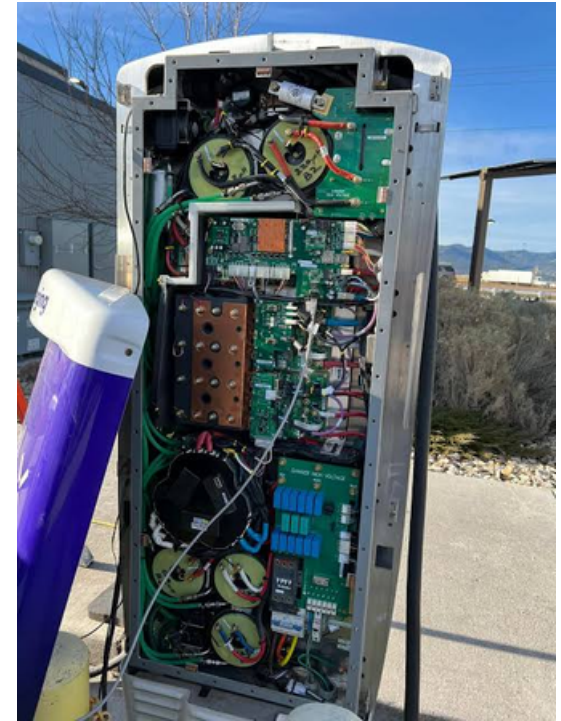
What Do We Usually Think About EV Charging ?

- An EV has one huge battery.
- The battery is just 12 V.
- A fast charger simply supplies more current. The battery is charged as one unit.
- The charging circuit is inside the EV



What Is EV Charging Actually

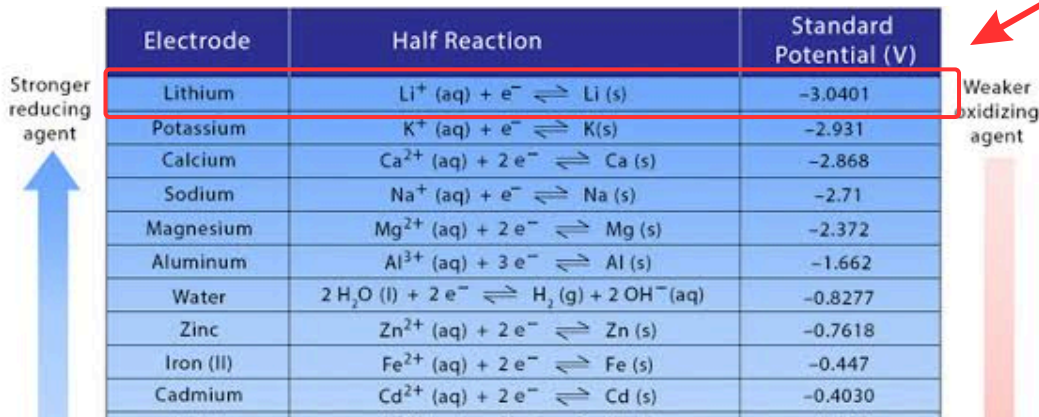
- EV battery is the combination of multiple small cells
- EV Batteries Operate at High Voltage (≈ 600 V)
- Main charging system is outside the EV in charging station



Why Not Use One Giant Battery Cell ?

The operating voltage of a lithium-ion cell is fundamentally constrained by the electrochemical potential of its electrode materials and the stability window of its electrolyte.

Electrochemical Series



Electrode	Half Reaction	Standard Potential (V)
Lithium	$\text{Li}^+ (\text{aq}) + \text{e}^- \rightleftharpoons \text{Li} (\text{s})$	-3.0401
Potassium	$\text{K}^+ (\text{aq}) + \text{e}^- \rightleftharpoons \text{K} (\text{s})$	-2.931
Calcium	$\text{Ca}^{2+} (\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Ca} (\text{s})$	-2.868
Sodium	$\text{Na}^+ (\text{aq}) + \text{e}^- \rightleftharpoons \text{Na} (\text{s})$	-2.71
Magnesium	$\text{Mg}^{2+} (\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Mg} (\text{s})$	-2.372
Aluminum	$\text{Al}^{3+} (\text{aq}) + 3 \text{e}^- \rightleftharpoons \text{Al} (\text{s})$	-1.662
Water	$2 \text{H}_2\text{O} (\text{l}) + 2 \text{e}^- \rightleftharpoons \text{H}_2 (\text{g}) + 2 \text{OH}^- (\text{aq})$	-0.8277
Zinc	$\text{Zn}^{2+} (\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Zn} (\text{s})$	-0.7618
Iron (II)	$\text{Fe}^{2+} (\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Fe} (\text{s})$	-0.447
Cadmium	$\text{Cd}^{2+} (\text{aq}) + 2 \text{e}^- \rightleftharpoons \text{Cd} (\text{s})$	-0.4030



Due to very high current the battery produce significantly very high amount of heat

Because of low voltage we need higher current for high power which increase the losses

$$P = I^2 R$$

Example: What if an EV Used a 12 V Battery?

Assume a **3S Li-ion battery pack**:

- 3 cells in series
- Nominal cell voltage = **3.7 V**
- Pack voltage:

$$V_{\text{pack}} = 3 \times 3.7 = 11.1 \text{ V}$$

Now suppose we want to run an EV motor requiring **30 kW**.

1. How much current is required?

Electrical power:

$$P = VI$$

Therefore:

$$I = \frac{P}{V}$$
$$I = \frac{30,000}{11.1}$$

$$I \approx 2703 \text{ A}$$

☹ The battery would need to supply approximately 2,700 A continuously!

$$R = 5 \text{ m}\Omega = 0.005\Omega$$

$$P_{\text{loss}} = I^2 R$$

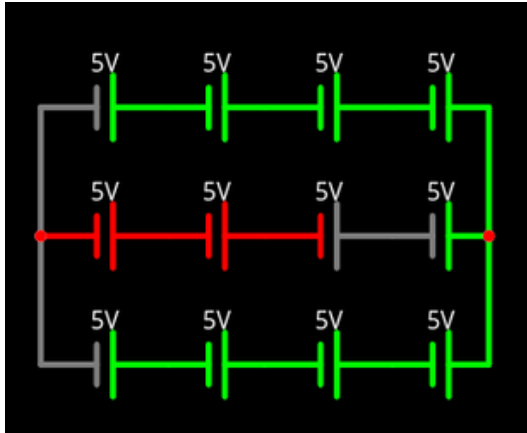
$$P_{\text{loss}} = (2703)^2 (0.005)$$

$$P_{\text{loss}} \approx 36.5 \text{ kW}$$

**Extremely inefficient
and thermally
impractical**

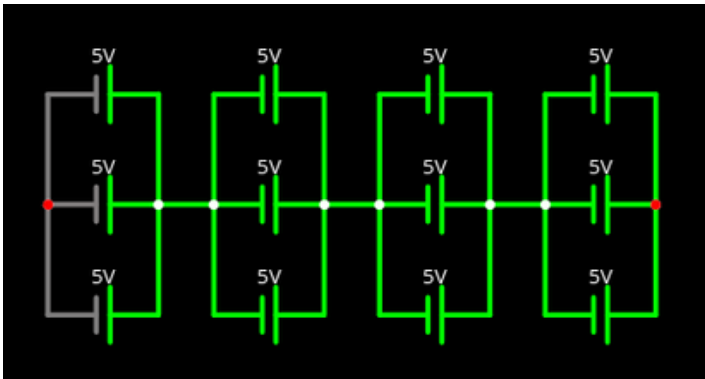
Topology of EV battery packs -

1. First series then parallel



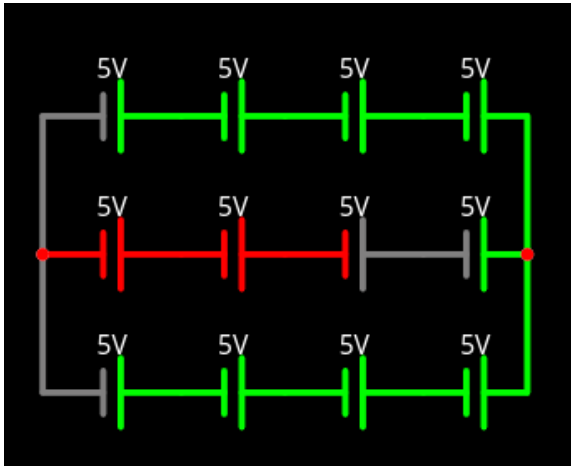
**Both has same
voltage and current
rating**

2. First Parallel then series



But still the most of the EV battery pack uses the First parallel then series combination

Problems with the first series then parallel combination -



- First series then parallel need more complex BMS design to control the voltage of each cell
- If a cell become short circuited then it create the voltage imbalance between strings
- A weak cell limits the entire series string

Cell 1 → Cell 2 → Cell 3 → ... → Cell 107 → Cell 108



weak/aged cell

Example -

String A: 108 cells in series → 402 V

String B: 108 cells in series → 398 V

String C: 108 cells in series → 400 V

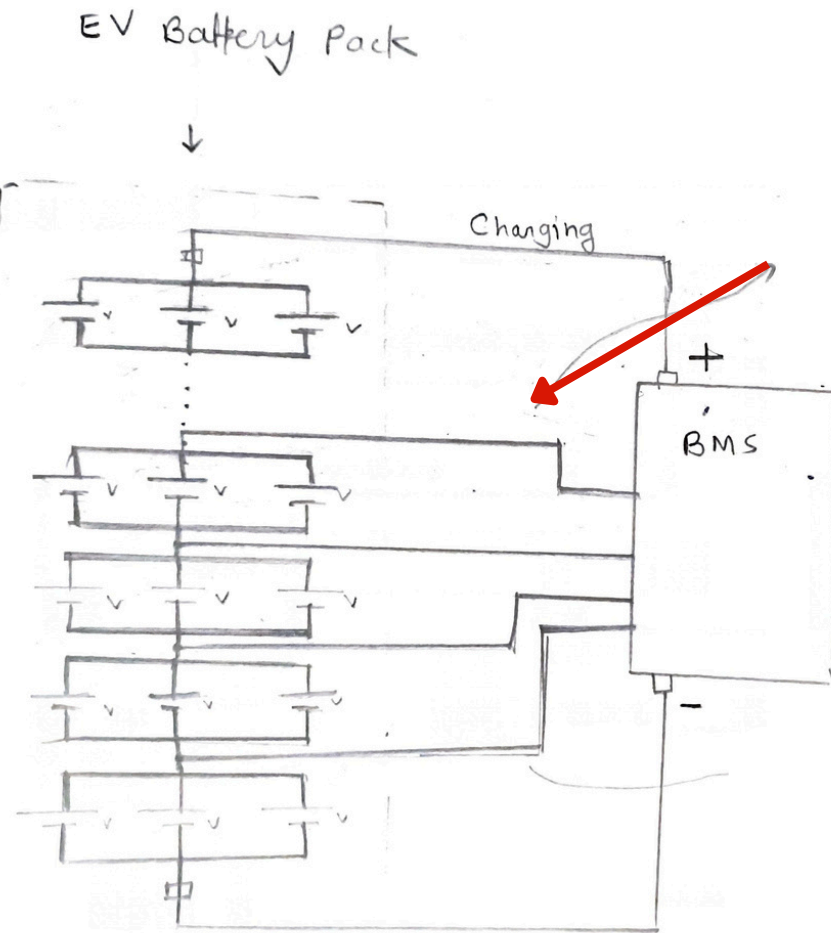
String D: 108 cells in series → 397 V



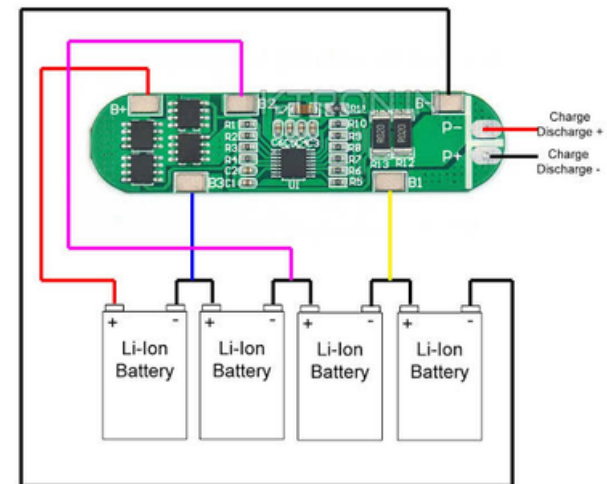
Large equalization currents can flow between the strings because they are at different voltages.

Battery Management System (BMS) Of EV -

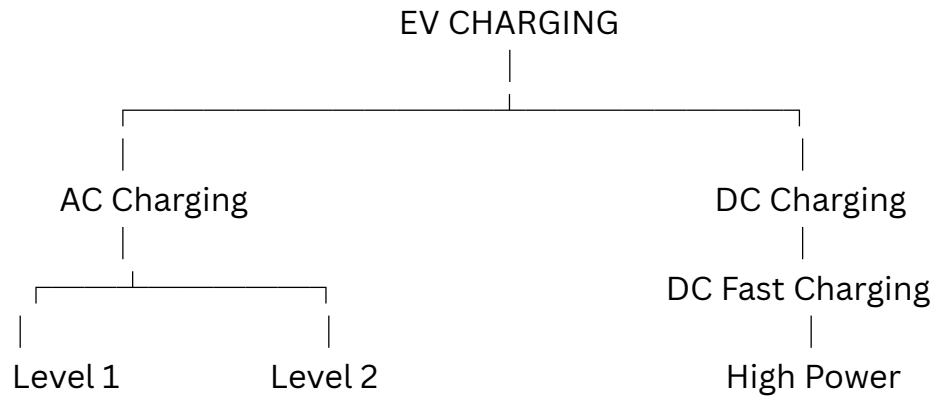
A BMS is an electronic system that monitors and controls a battery pack by managing parameters such as voltage, current, temperature, and state of charge to ensure safe, efficient, and reliable operation.



These are use to measure the voltage and bypass the charging current when a particular set of battery



EV Battery Charging System -



AC Charging -

Grid AC → EV onboard charger → DC → Battery

Level 1 - Single phase charging

Level 2 - Three phase charging

DC Charging -

Grid AC → external charger → controlled DC → Battery



Level 1 charger

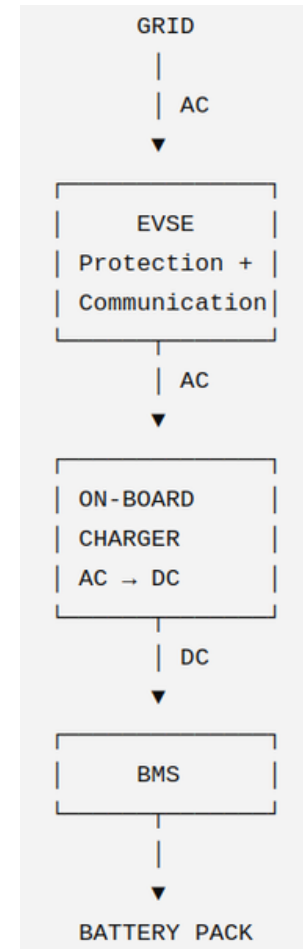


Level 2 charger

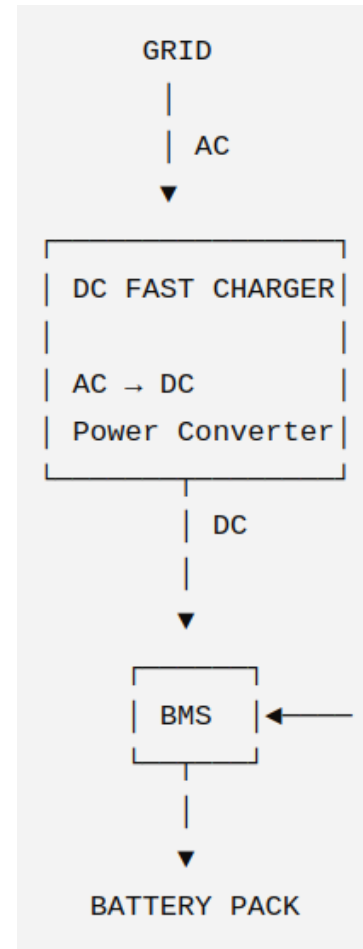


DC Fast Charger

AC charging architecture



DC charging architecture



Level 1 Charging

$$P \approx VI$$

For example:

$$V = 230V, \quad I = 16A$$

$$P = 230 \times 16 = 3680W$$

$$P = 3.68 \text{ kW}$$

Charging time

Suppose:

**Very low
power**

$$\text{Battery} = 60 \text{ kWh}$$

and charger power:

$$P = 3.68 \text{ kW}$$

Ideal charging time:

$$t = \frac{E}{P}$$

$$t = \frac{60}{3.68}$$

$$t \approx 16.30 \text{ h}$$

**Take very long time to
charge**

Level 2 Charging

Level 2 is typically much higher-power AC charging.

For single-phase:

$$P = VI \cos \phi$$

For three-phase:

**Using three
phase
system**

$$P = \sqrt{3}V_{LL}I \cos \phi$$

This is an excellent opportunity to explain **why three-phase charging is attractive**.

For example, suppose:

$$V_{LL} = 400V$$

$$I = 32A$$

$$PF = 0.99$$

Then:

$$P = \sqrt{3}(400)(32)(0.99)$$

$$P \approx 21.9 \text{ kW}$$

Now compare that with a 230-V single-phase supply:

$$P = 230 \times 32 \times 0.99$$

$$P \approx 7.29 \text{ kW}$$

**Generate significantly more
power**

DC Charging

Therefore:

$$P = VI$$

$$I = \frac{P}{V}$$

Suppose we want:

$$P = 200kW$$

At 400 V

Decreasing
current by
increasing
the voltage

$$I = \frac{200000}{400}$$

$$I = 500A$$

At 800 V

$$I = \frac{200000}{800}$$

$$I = 250A$$

Same power, half the current.

Now introduce conductor loss:

$$P_{loss} = I^2 R$$

Therefore:

$$P_{loss} \propto I^2$$

If current is reduced from 500 A to 250 A:

$$\frac{P_{loss,800}}{P_{loss,400}} = \left(\frac{250}{500}\right)^2 = \frac{1}{4}$$



Level 1 Charger



Level 2 Charger



DC Charger



Source: <https://www.youtube.com/watch?v=jZBsOud409Q>

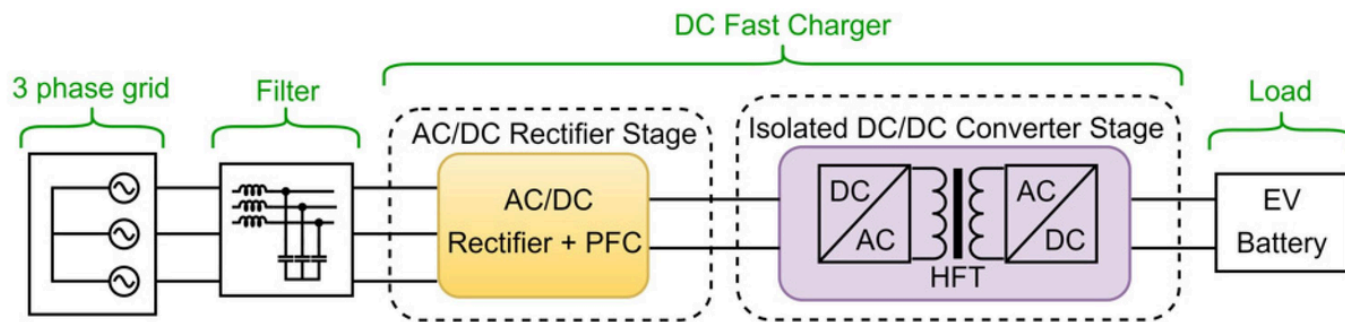


Increasing the voltage
twice reduce the power
loss by 1/4th

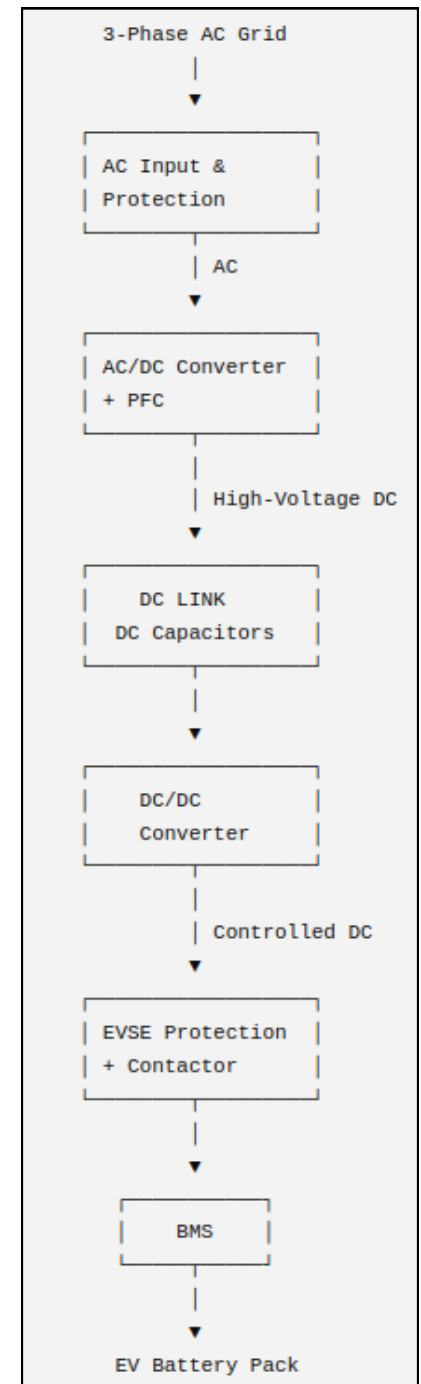
Comparison table -

Parameter	Level 1	Level 2	DC Fast
Supply	AC	AC	DC at vehicle
Conversion	On-board	On-board	Off-board
Typical application	Home	Home/work/public	Highway/public
Power	Low	Higher	Very high
Charging current	Low	Moderate	High
Charging time	Long	Moderate	Short
Grid impact	Lower per vehicle	Higher	High
Thermal management	Moderate	Moderate	Critical

DC Charging Topology -

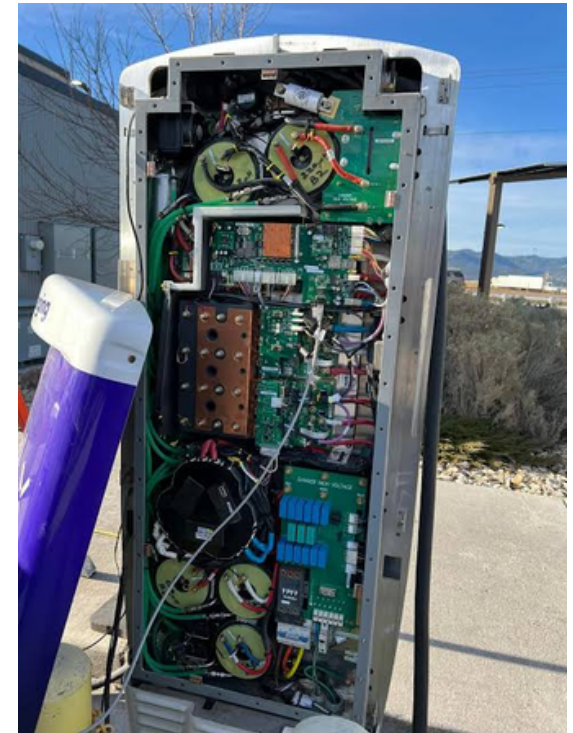


- First we convert the AC into DC and then we use high frequency DC to DC converter to step up the voltage
- we convert the first AC to DC then step up the voltage because the EV battery voltage will vary too much
- And due to high frequency switching we need small ferrite core transformers instead on large iron core transformer which create huge loss



Why fast EV Chargers are very big in size -

- They handle hundreds of kilowatts of electrical power.
- **High current causes significant I^2R losses.**
- Large heat sinks, fans, or liquid-cooling systems are needed.
- Large rectifiers, switches, inductors, and capacitors are required.



Think about a 200 kW DC fast charger:

$$P = 200 \text{ kW}$$

Even if the charger is 95% efficient:

**These energy losses
create lots of heat**

$$P_{\text{loss}} = P (1 - \eta)$$

$$P_{\text{loss}} = 200(1 - 0.95)$$

$$P_{\text{loss}} = 10 \text{ kW}$$

That's 10 kW of heat that has to be removed continuously!

Thank You ! :)